

EEE 120

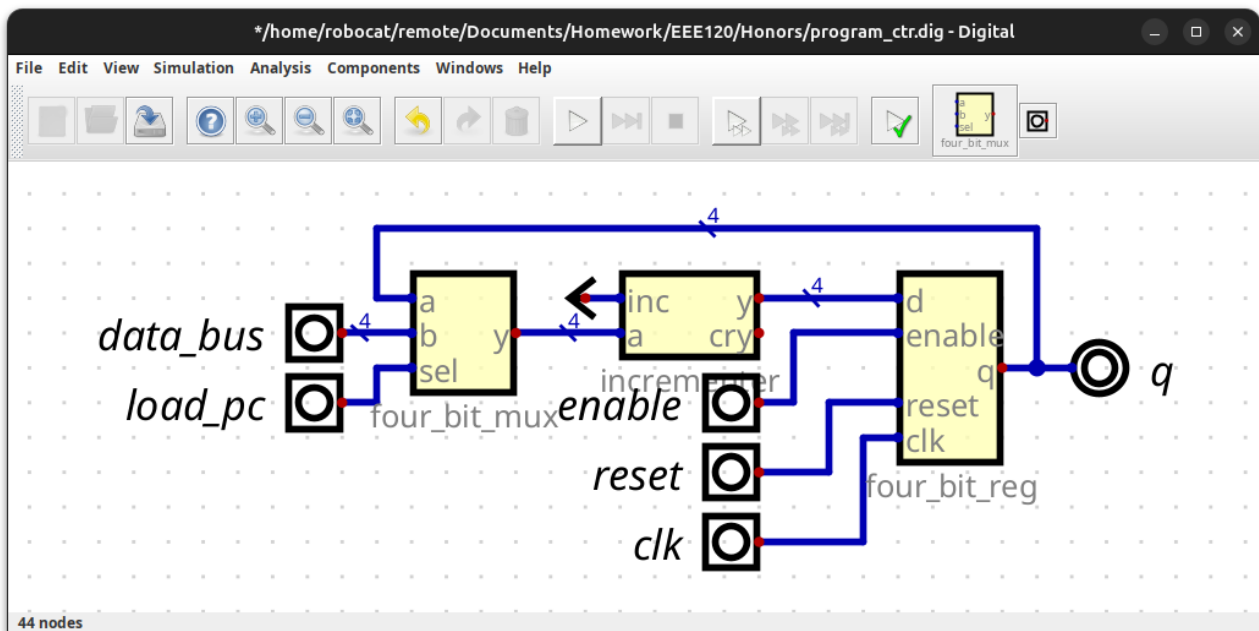
Honors Lab Answer Sheet

Name: Dylan Shah

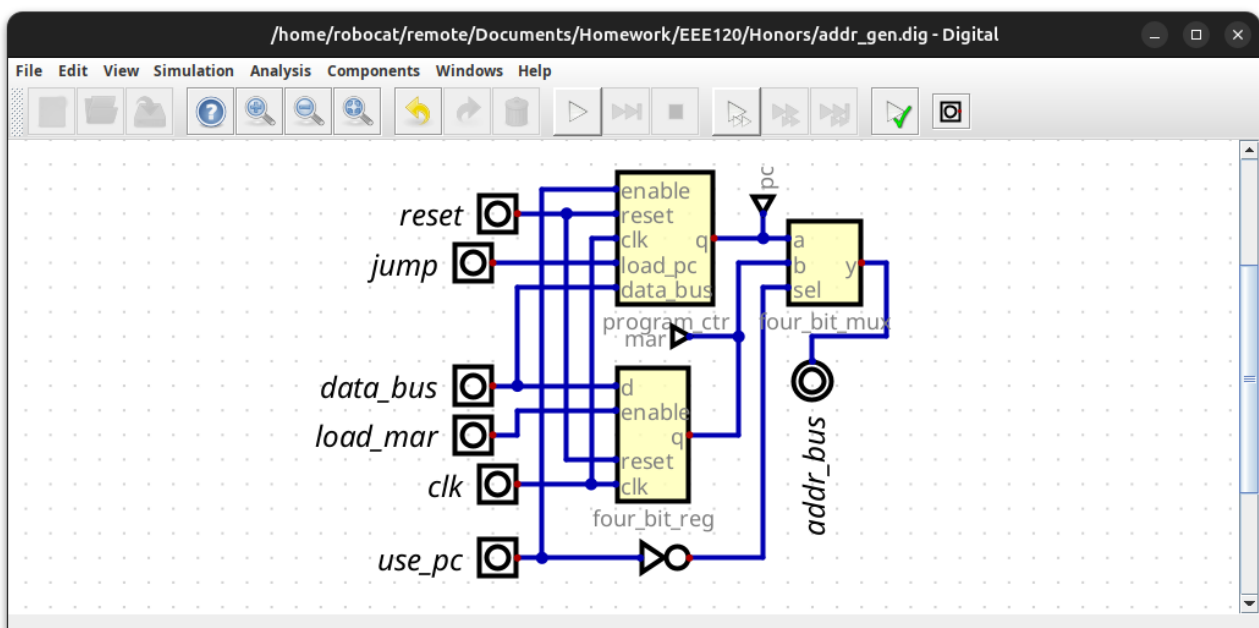
Instructor/Time: Dr. Steve Millman Tu Th 3:00 PM - 4:15 PM Date: March 26, 2026

Task H-1: Add the JUMP Instruction

The updated program counter:

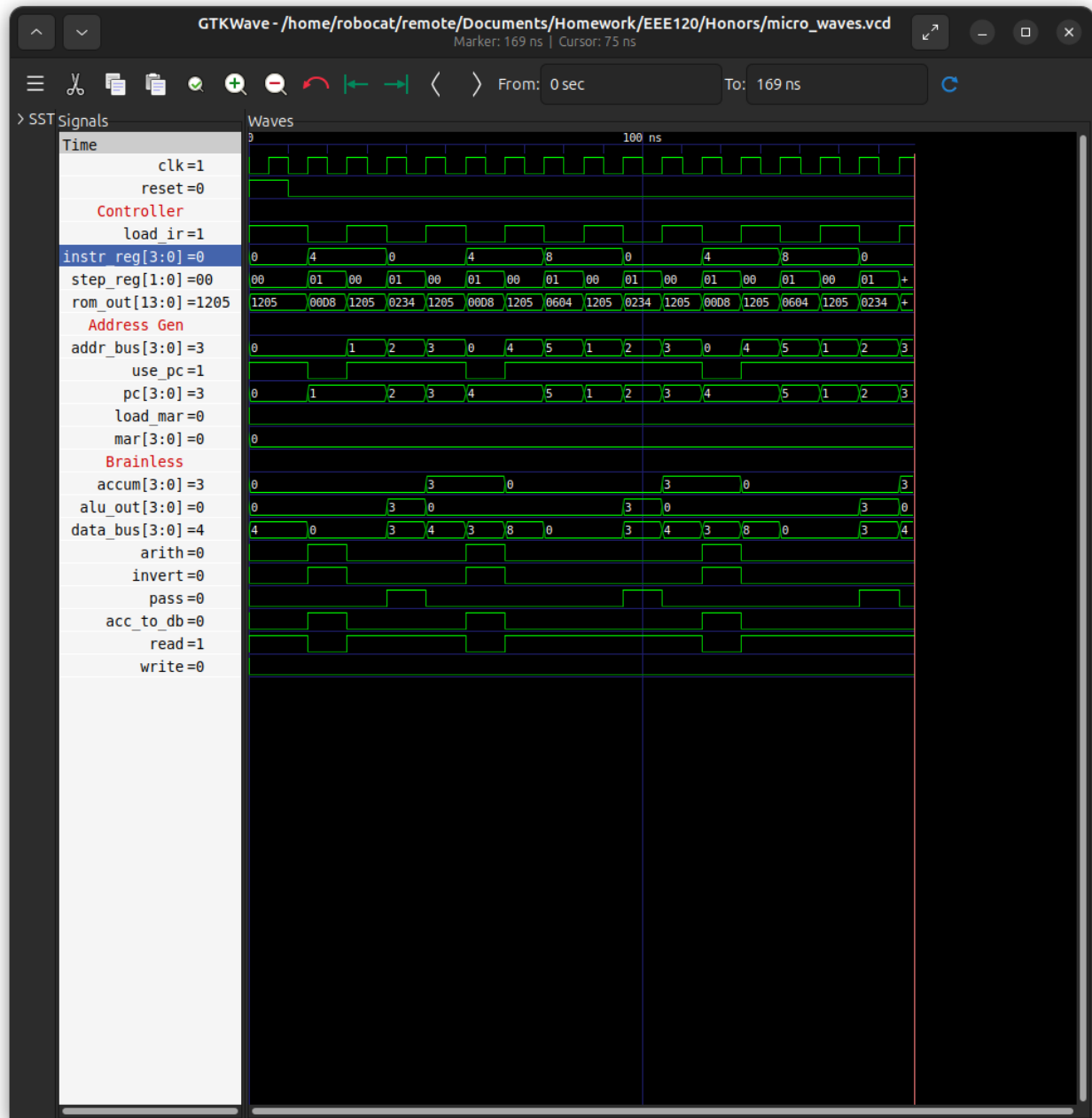


Updated Adder_gen:



Updated Microprocessor:

3
 4 // Zero
 8 // Jump
 0 // to load
 2 // Stop

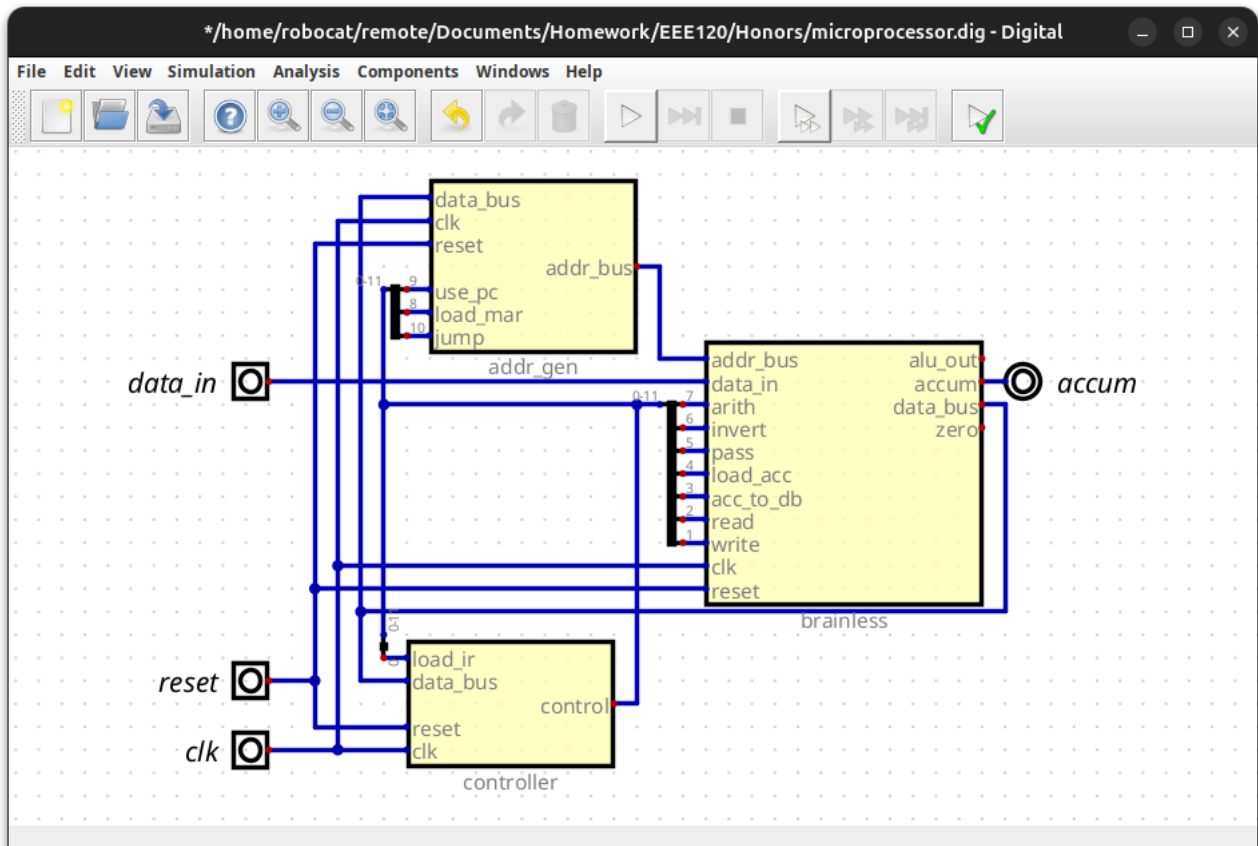


Challenges I faced:

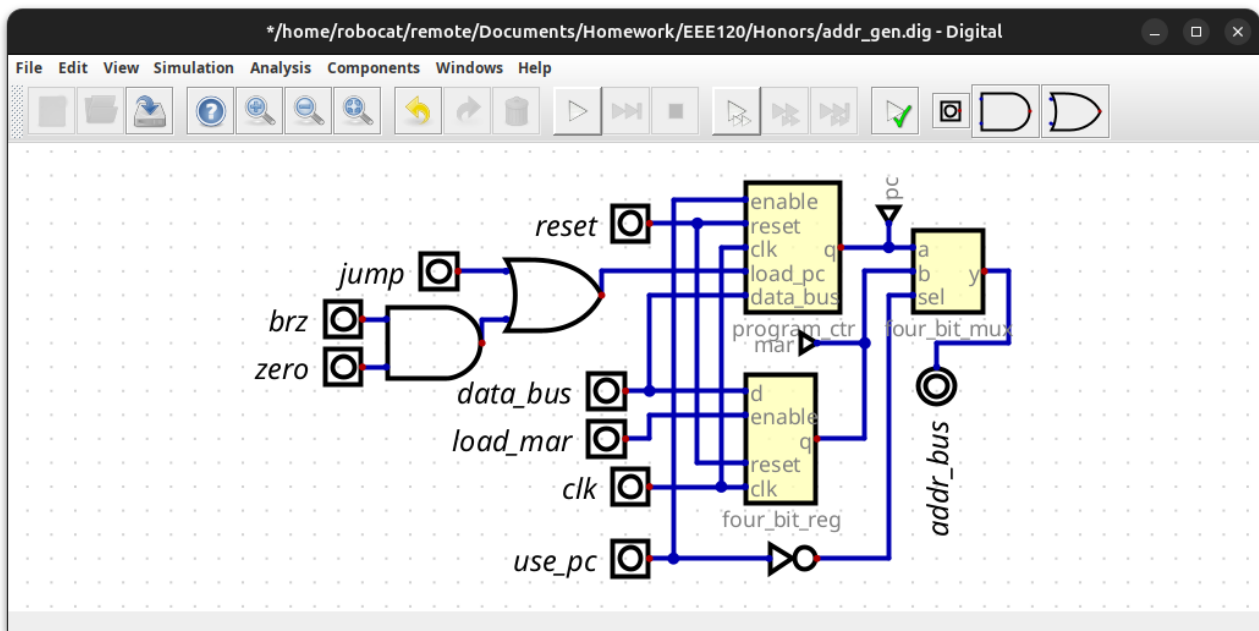
I was setting the MAR insted of the PC with the microcode at first, it took me a while to see the issue as you can not see inside of components in Digital.

H2: Add the BRANCH ON ZERO Instruction

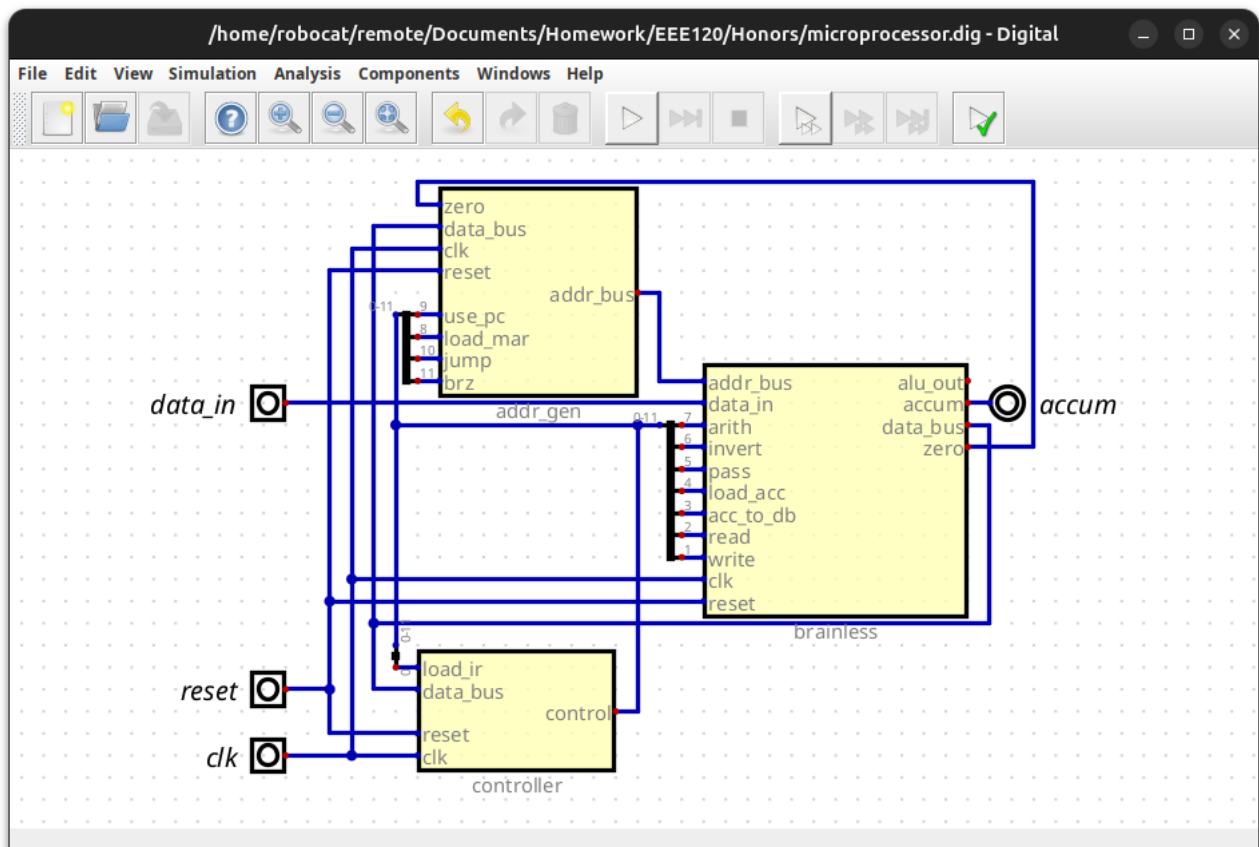
Updated Brainless:



Updated Adder_gen:



Updated Microprocessor:



Updated ROM:

...

1205 # Jump if zero

0A04

0000

0000

RAM: (One with loading Data In)

7 // Load ACC with Data In

5 // Subtract one

1

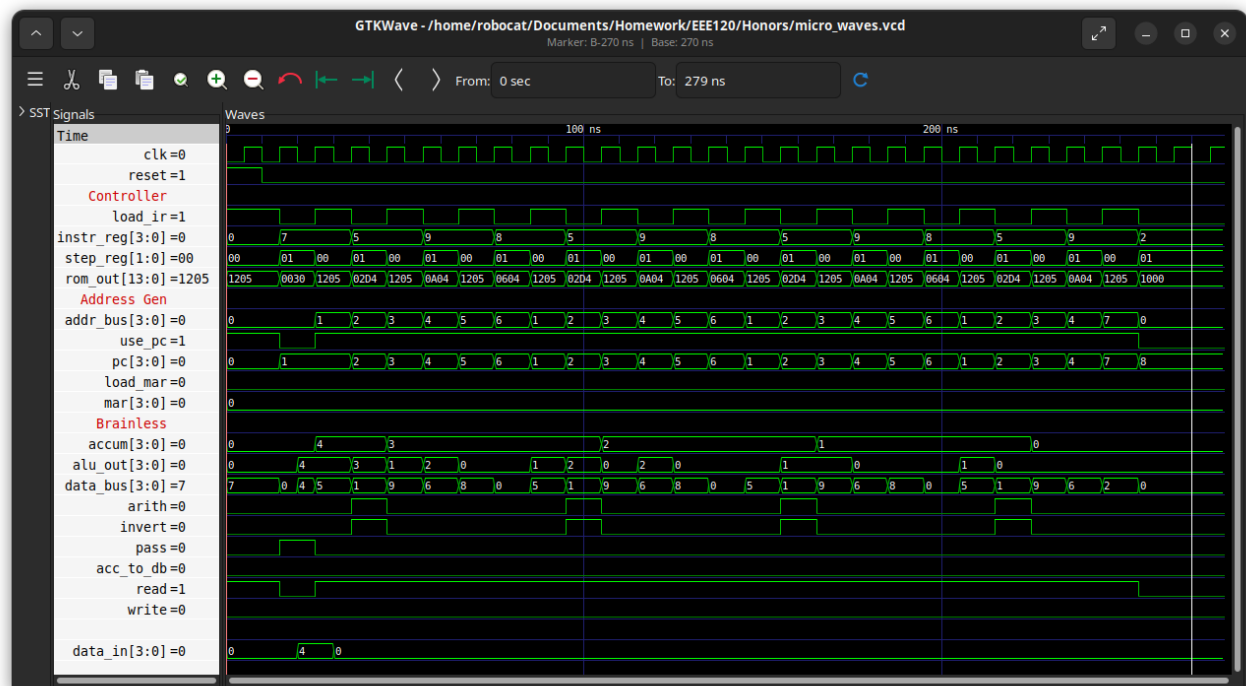
9 // Break loop if zero

6

8 // Jump to the subtract

0

2 // Stop



RAM (Given):

0x0 # Load ACC

0x1 # a nonzero value

0x9 # BRZ instruction

0x9 # not taken since accum is not 0

0x0 # LOAD ACC

0x0 # zero!

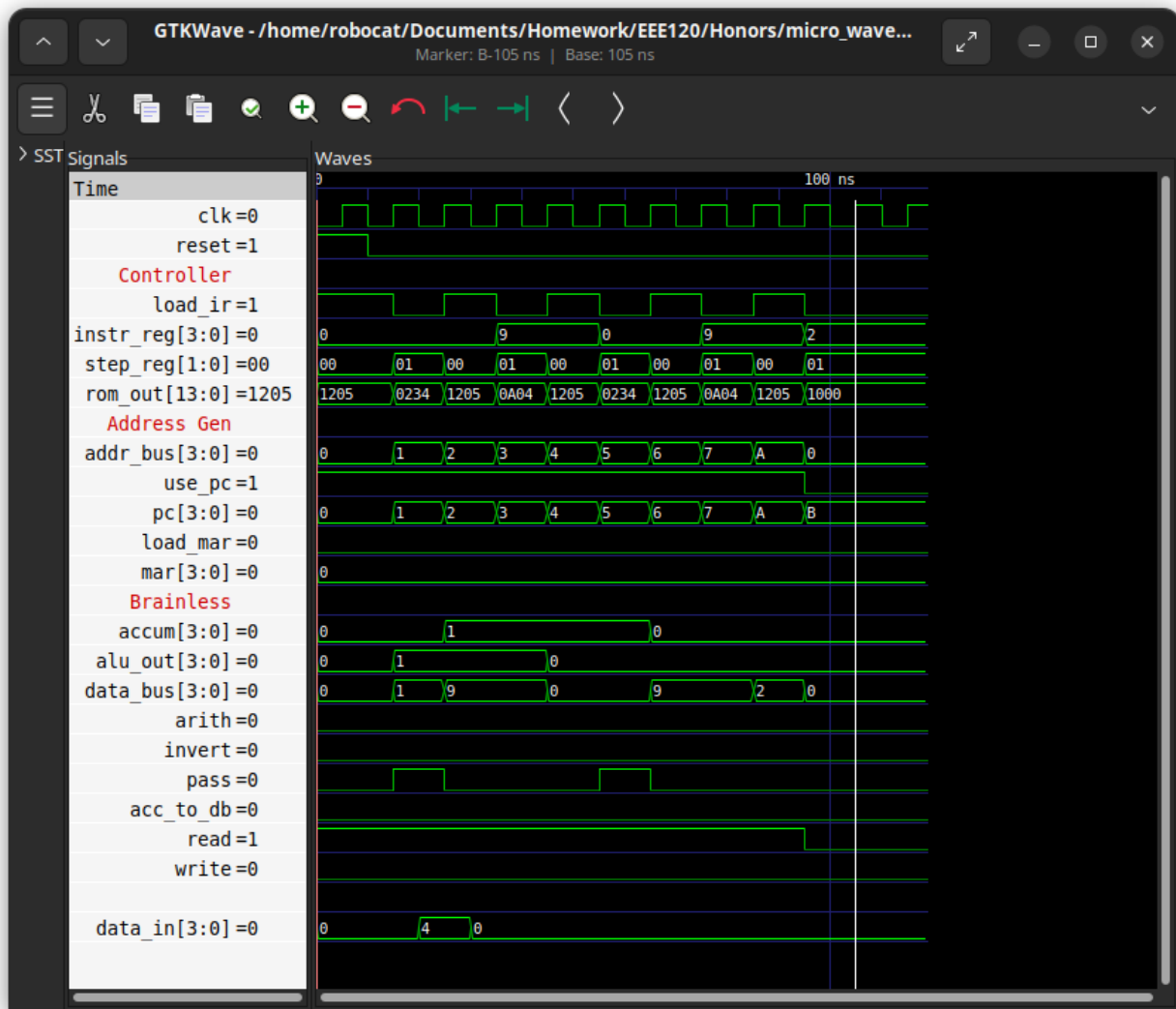
0x9 # BRZ instruction

0x9 # branch to address 0xA

0x0 # Load ACC

0x5 # this should be skipped!

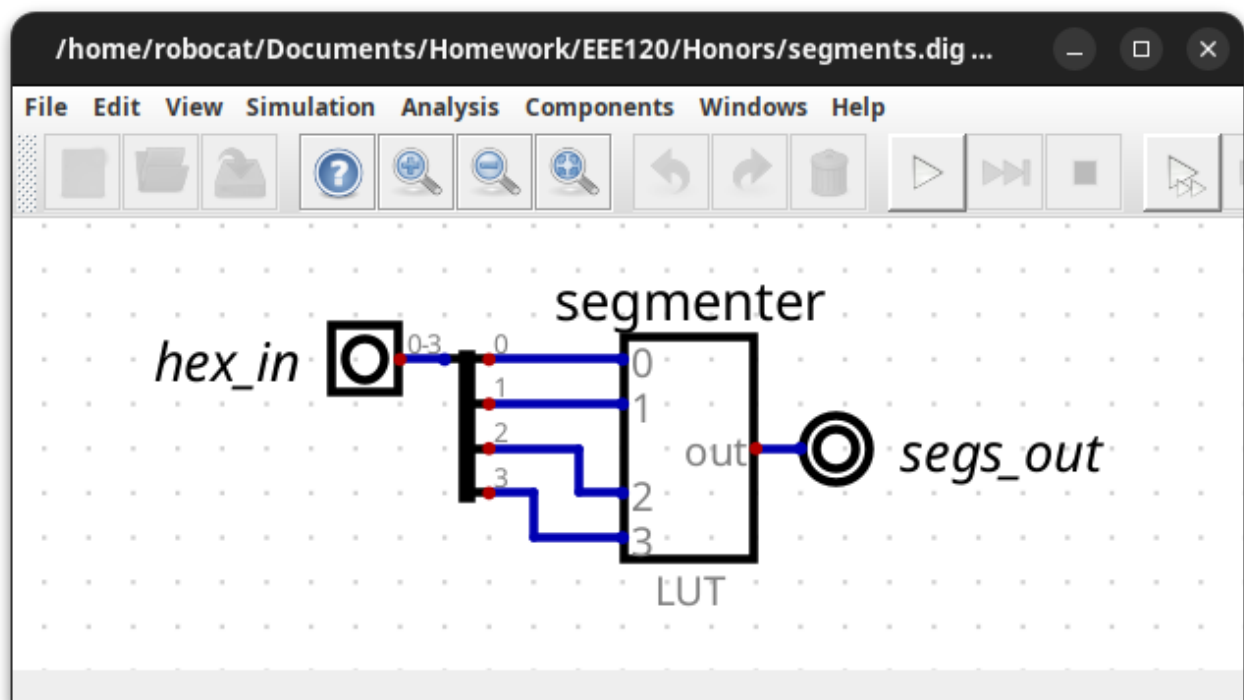
0x2 # STOP



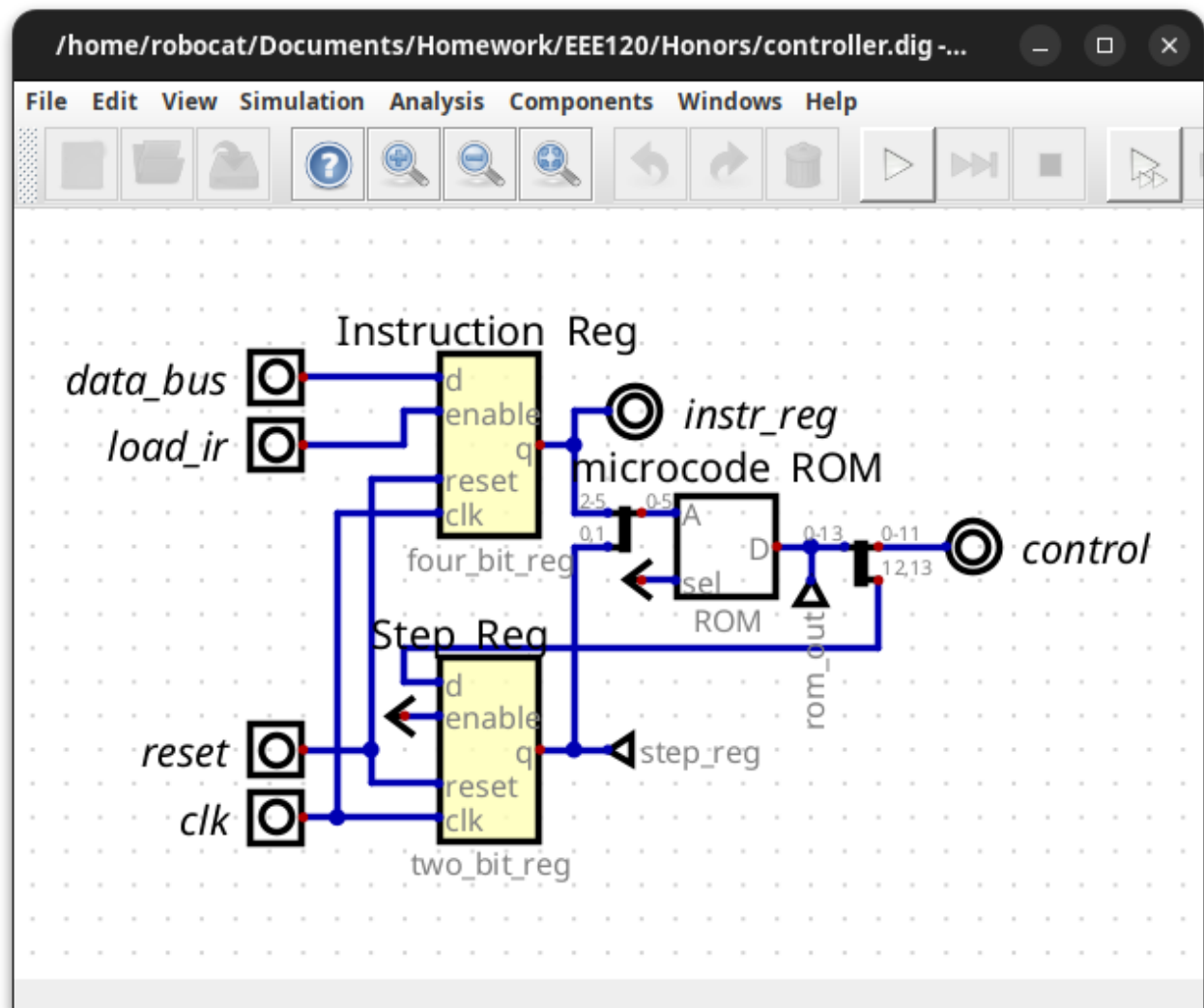
Challenges I faced:

Somehow, in the rom, the setting to preload at start was selected and it was loading my values from Lab 4. IDK how this happened or why the previous one worked, but that was fun to debug.

H-3: Build the Seven-Segment Display Translator



H-4: Bring the Instruction Register to the Top



H-5: Invert the Reset

